Technology Stack

Date	10 July 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyzer
Maximum Marks	2 Marks

Define Your Technology Stack

The Voice-Based Concept Understanding Analyzer is developed using a modern, scalable technology stack consisting of frontend technologies, backend frameworks, machine learning libraries, databases, deployment platforms, and external APIs. These technologies were selected for performance, maintainability, flexibility, and ease of development.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	HTML5, CSS3, Streamlit	Provides a responsive user interface with fast development and an interactive web experience.

2	**Backend / Server-Side**	Python	Handles speech processing, AI integration, report generation and application logic efficiently.
3	**Database / Data Storage**	JSON Files, Local Storage	Stores reference concepts, reports and application data in a lightweight format.
4	**Cloud / Hosting / Deployment**	Streamlit Community Cloud	Provides free deployment, easy updates and cloud accessibility.
5	**Version Control & CI/CD**	Git & GitHub	Supports version control, collaboration and project management.
6	**Third-Party APIs / Other Tools**	OpenAI Whisper, Sentence Transformers, Librosa, FPDF, FFmpeg	Used for speech transcription, semantic similarity, audio feature extraction, PDF generation and audio processing.